

ActInf Textbook Group ~ Cohort 4 ~ Meeting 9

(Chapter 4 part 1)

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hi everyone it's August 1st and we're in cohort four having our first discussion on chapter four so before we go into more detail on chapter four and everything does anyone have any just overall thoughts or or comments on chapter four what what did they uh expect it would be delivering as a chapter how how was it or what was any aspect that that people remembered or anything um I uh found this chapter to be rather uh mathematically dense I suppose and the authors do warn of that at the beginning of the chapter um I mean I think it's great so far as it starts to introduce how they're using Bayes theorem how they kind of exploit Jensen's inequality to move in the direction of like calculating uh free energy [Music] using logarithmic functions in order to do that you know the initial models of the static perceptual tasks also seems like a really nice introduction moving on to bomb DPS and as well as the continuous models used in active inference so in a in a way it felt like it uh kind of started with nice like first principles and built up from there but that said I can start to see maybe the necessity of having a math learning group for uh some folks here given this uh chapter thanks yeah anyone else any like just overall thoughts or well I think yeah I would just totally agree 100 with what Andrew shared basically it's just similar thoughts I mean basically the the maths uh are very well explained and it goes gradually but I think it would be beneficial here to yeah to go even deeper and slower through them maybe separately definitely any other thoughts people have yeah Darius so then anyone else yeah Daniel it's less a point about the chapter although I agree with the previous sentiments that is mathematically dense it's more of a point about the mass learning group I think myself and several others just having trouble locating it in terms of where it actually is on the Discord so the past couple of weeks I've been trying to get into some of the voice chats and it doesn't there doesn't seem to be anything kind of happening on them so there might be a bit of confusion yeah thanks well it is uh people can use the general voice chat just the top voice Channel and uh you know if there's no one there just still jump in and stick in there okay because otherwise I see sometimes people jump in and then

immediately jump out and then that then other people think that there aren't anyone around so just jump in and hang in and you'll be able to see people's like icon and see what channel they're in I mean yeah it's a it's a it's a decentralized learning group what I people can edit Pages or or just ping me if there's any support needed I'll try to join what I can but it can I mean there's there's so so much work on the more introductory side like kind of just going from the fundamentals on through all all the work that Jonathan and Ali and others are doing too so there's no no lack of things to do in math learning group I hope people do like connect on it um Young wait I don't hear you I can see you're unused baby Frozen though can anyone else hear it or no okay okay when you're when you're back feel free to continue any other um General thoughts on four or anything writing in the chat just another small specific observation that uh as a kind of a practicing entry-level data scientist uh the way something that helped me understand the static perceptual models earlier on is that it almost feels like it's just describing something like the process of binary and classification yeah one of the models and uh figure 4.2 the very first one you have the example of like uh your your prior being a prevalence of a disease is there a disease yes or no you have uh your true positive rate your true negative rate and you end up with the result of a new test I found that to be rather interesting and useful just are we trying to predict does someone have a disease or not there's more that goes into that step by step yeah thanks that's that's one of the classic uh Bayesian um examples like there's a test that's 99 accurate and there's a disease that one in ten thousand people have what is the probability that you have the disease if you got a positive test result and the kind of snap answer type 1 answer is um 99 because if it was positive and it's 99 accurate then how could it not be 99 but when you actually unpack the base equation it's actually not 99 percent it's actually still unlikely um in that with those parameters um young Hoon oh yeah sorry about that uh so I I find these mathematical equations in chapter four to be a little bit ambiguous especially like uh the notations or like Dimensions or whether it's a vector or whether it's a scalar or Etc depending on the context like I find especially uh difficulty to understand the exam equation when there are some functions called like categorical function so I I just accepted the concept but um just wanted to make it more verified so I actually uploaded some PDF and questions to the chapter four because I I got stuck with some um derivations of it so that might be in the chord or chapter four questions and you can look it up later on awesome awesome we'll definitely look at it and also here I see one this overleaf link this was prepared wow I guess he just added this in this is from Jonathan shock yes I found it to be very helpful that's that's awesome thank you Jonathan yeah I mean

there's even already there's all there's so many pieces of material it's it's um you know it's like a second level question how we how we organize things so that they're discoverable and kind of annotated so that somebody who's looking for like this level of detail can connect with this type of material let's just read his basic remarks free energy and expected free energy are introduced almost in parallel but it's not clear which one should be used when or exactly why one needs both of them this is relatively simple that the free energy is used to improve the model itself whereas the expected free energy is used to make decisions about which is the best action to take which will lead to states which are preferable on states which lead to more epistemic value which will themselves themselves allow one to update the free energy yeah this looks really good any other like just thought or question or area in four that people want to jump to yeah young again oh yeah so I'm I'm not sure whether it's a good explanation or I'm not sure whether it's a correct intuition but for me it feels like um like comparing with reinforcement learning uh reducing the free energy is something a little bit related to offline learning while minimizing the expected free energy seems like to be related to online learning so you choose these decisions and then you try to reduce the free energy online or the expected free energy while you reduce the variation of free energy offline so I'm not I'm not saying that those are equivalents but it seems to me that they are somehow related maybe just summarize what is online oh uh not really a master of reinforcement learning but uh based on what I understood about it is that online learning is learning while you're at doing the action while offline learning is something more like you learn the uh value of something based on the state trajectories that you already visited so it's more like learning week by replaying I would say it can if anyone thinks that it's I'm explaining it in a very ambiguous term or if I'm wrong just let me know cool yeah you're definitely an area to look into Darius I think what might be useful for me and perhaps others is getting a bit more clarity on exactly how we Define the generative model in Acts of inference so on one side um and you hit and you see it expressed in this chapter that it comprises the prior and the likelihood term in base's equation um but then on the other I've you know seen other theorists talk about how it's actually basically the entire um conditioning dependent Dynamics across the Markov blanket so it includes the external Dynamics it's not just what's internally encoded in the internal states of the particle so unless that's just a continuing conflict A continuing debate within the active inference Community it would be good just to get a little bit of clarity because I think that also helps us explain exactly what is minimizing free energy and whether we can talk about an agent having a generative model or being a generative model foreign yes it's a great point and it has come

up before so in fact there are about one one and a half definitions of what the generative model is and they're um they're not identical they're kind of nested Concepts and and it certainly does matter and so I can understand why it's a point of confusion um in the 2022 textbook that we're reading right now generative model is very very clearly differentiated from the generative process so like when we're looking at the figures it's like here's a generative model that's like the agents sometimes we would call that the particular States because it it constitutes the whole particle like just like if we were talking about a dust particle Brownian motion now we're talking about the cognitive particle so that includes the blanket States and the internal States and so that is sometimes called a generative model and the generative model as it's framed in the textbook is to understand or to make this model of the generative process which is the actual process that gives rise to observations and so this distinction between generative process and is um is heavily used in the textbook however more recently some authors have have uh taken a slightly difference approach or definition so as you mentioned and for example in in this representation from Maxwell on um I guess a month ago or something around then here generative model is used to describe the entire particular partition all the States including the external States um and in and here's all these questions earlier there was an emphasis on distinguishing between generative process and generative model but in this figure figure one which you've shown the slides we don't see any generative process so where does generative process fit in the diagram what is the distinction between the concepts and yeah all all put these into the coda as well at least Maxwell is not using the generative model generative process distinction as much here's how I kind of reconcile this whatever you want to call different parts of the particular partition you need all four of them to be defined in the model so whether you just call them you know whether you uh buy them a la carte or whether you buy them in a bundle all the particular States and the all the states need to be described um otherwise you're going to have a null variable um and now there's several um we we could you know we could talk about uh you know you have your friends who like this music and you have your friends who like this food and so there there's there's different nested levels of describing the French group with these four friends one of them is the way that the textbook describes with the generative process external only and the generative model with these three on the right blanket and internal another um group that we talk about sometimes is just the blanket States of course states are not intrinsically blanket States because every state is blanket is something else and so on but in this setting blanket and then another um set of states that we talk about sometimes are the autonomous States which is just the internal States and the active

States autonomous states are the two things that can be changed to reduce free energy change your mind learning perception attention Etc and action change the world so that's the no thumb on the scale so whether or not you focus on these three on the right and call it a generative model and this is a generative process or whether you call all four the at the end of the day the variables are defined the same way and the variational and the expected free energy are still regarding the autonomous states with respect to the sensory States and so generative model and generative process is a little bit of like a linguistic descriptor or handle but it it it doesn't change anything to give a different English name to describe a subset of notes so I don't think it's an issue however it is confusing well I just got one more point of clarification on that yeah yeah you go back to that diagram that would be quite useful actually here so yeah that's perfect um in terms of self-evidencing then if we take Maxwell's idea that it's degenerative model is the system the across the Markov blanket do we talk could one consider that system to be self-evidencing or is it only the particle which excludes the external states which is self-evidencing that's a great question I think also calls back to your um rock climbing example okay the rock climber is uh doing something cognitive about the wall but is the wall doing something cognitive about the climber or is there some are we considering the entire system as doing some kind of gradient descents yeah that's it's a good um question either side of the blanket I believe may be seen as doing some kind of self-evidencing now in the case of one active adoptive sentient entity and one inert one the inert entities self evidencing is going to be um trivial so just because there's um self-evidencing on both sides of the mic it doesn't mean it's the same kind of self-evidencing but you can also make situations where it is the same kind of self evidencing across both sides like in the bird song example does that uh do two or uh when when two sentient agents are engaged in self-evidencing does that constitute higher uh level of self-evidencing at the entire system level description possibly I I mean if if um if they're both reducing free energy now what if one is um reducing free energy and that is creating an increase in free energy for the other could that increase the net free energy on the system I I don't know or is it always the case that that it um that the ball has to roll downhill and the more massive cognitively massive self-evidencer takes a stronger hand and and that makes it move down on average also I don't know Andrew I don't wanna pull too far away from the kind of current formulation that's being discussed here but something that um has been useful for me and it I think I please correct me if I'm misstating this but uh with differentiating the process and part is that the generative model is presumed to have um something like preferences or the the its own C Matrix or its own homeostatic set points which I believe the

generative process does not have so we have like from chapter three and so on and the generative process or the environment of the the agent the generative model is presumed to have like you know dissipative forces entropy and so on and so the generative model is kind of acting against those in order to realize its own uh expectations or preferences that is uh I I want to survive as a generative model the environment around me has no set of direct preferences saying like Andrew needs to survive I'm the one who's saying that and so despite the fact that I'm attempting to uh infer or approximate the external States so that I can get around in the world like I'm also kind of acting against it or or at least you know in a way that doesn't it's never going to exactly equate to the the generative process around me and this is where the map does not equal the territory uh kind of uh formulation starts to come into play and becomes all the more is any of that track yeah that that's a it's a great um points um certainly map doesn't have to be the territory even structurally like the generative model too cold just right too hot three state categorical that's the categorical distribution or it could be um uh continuous variable zero to one or zero to 100 and then outside it could be a totally different structure and so the generative process I I believe it or if you just call it you know the process generating observations that could just be a sine wave or it could be a square wave or it could be a deterministic um system and so that that makes it very clear like like you said that when we're talking about the generative model of the agents we're talking about a b c d e those are the variables that have to be defined that's figure 4.3 basically what whether we're working in continuous time or in um uh discrete time these are the variables that have to be defined or counted for in the generative model this doesn't describe how those observations are generated so this is kind of like just the agent lifted from the environment and so it still is helpful and it's the one that's exerting preference and engaging in sentient behavior and all of that and then it's like helpful again whether you call it generative process or underlying process generating the observations or the components of the system that generate the agent's observations just to make it clear that we're not always in a situation where it's two active entities talking to each other like the bird song example but rather we can be in a situation where um like the rock climber that Darius brought up where you have like this figure 4.3 agents and then there can be an articulation with a generative process again or whatever you want to call it that is not doing like policy selection except in a trivial way yeah these are all great points and and it's funny you know it reflects how things update and and evolve and helps us get a get a parallax on on what's really happening what are like just the purely verbal descriptors we're actually like the term like the English term isn't doing any work like generative model generative

process the term doesn't necessarily do any work what what does do work though is statements like this calculation of free energy requires data why family of variational distribution Q and the generative model those three things together can get a and that also helps reveal like why discussion of well when this happens then the free energy would go up those are speculative statements they can be super useful and interesting but until the calculation has occurred it would be like um talking about a linear aggression that hasn't been performed or talking about a time series analysis that hasn't been performed so it makes it clear where's the conversational and the speculative and then how do we get to a point where we can calculate the free energy about the data with respect to the distributions equation 2.5 variational free energy about the variational distribution and the data so that's how we get there any other kind of thoughts on four also Ellie and a few people probably have watched it but just a few days ago Ollie and I did a video overview on four well we can um turn to a question that someone has okay both these questions look very good and they're later slightly later in the chapter and slightly um with the math details um I wonder if Jonathan's work May address some of this but we'll try to get there let's start a little earlier um on and also yeah people can can um upvote the questions so that we can pick a place to start but um does anyone have a specific question they want to bring up or do we want to walk through like through and and just get to just see what the sections are about Bayesian brain describes using Bayesian statistics well there's sort of an instrumentalist flavor and a realist flavor as with many things the realist flavor would be the brain does Bayesian statistics which might be very deep might be very um overreaching the instrumentalist flavor is we can model what the brain is doing with Bayesian statistics which is to say like updating what kinds of Bayesian statistics well the inbound sense making signal processing side and then the outbound action selection control theory side two operations of kind of the general cybernetic entity and we can use Bayesian models Bayesian methods there um that's the problem of inferring states of the world perception inbound sense making and inferring the course of action action selection outbound planning exact inference is generally computationally intractable or infeasible separate from that limitation in psychology often what's being considered is bounded rationality or like constraint Limited cognition so how can we respect that we're modeling bounded rationality approximate rationality limited finite systems and kind of um at the same time um move Beyond this limitation of exact base being intractable um like in the uh frog jumping out of the hand example it was just some small matrices to multiply and so the exact base in calculation was not out of our grasp however if we were going to be doing some kind of large-scale data analysis of course we would need to

do some approximation methods there's two large just just speaking broadly there's two ways to approach this kind of approximate Bayesian uh calculation and they're related and there there's many many sub-flavors and live stream 52 is a good place to learn more about this but there's sampling methods and variational methods sampling based methods or Monte Carlo methods that came from the card games that were played in Monte Carlo and it was like well like what's the probability of getting um two twos uh four red cards and one clubs I mean you could write an expression and try to analytically solve that the UR the sampling based approach which has become called the Monte Carlo approach is where you just draw um hands of cards and then you draw a million hands and you say okay then there's three so then in the end you say my best estimate is this happens three out of a million times now you can probably see a lot of um like limitations but advantages of sampling on one hand it allows you to approximate arbitrary potentially analytically unruly so even where you don't have a nice smooth function or you don't know a lot about the system sampling may work quite well on the other hand you never really know if you're sampling appropriately or if you've sampled the completion the other methods that are utilized and also this cannot be done incrementally very well at all you need to curate many many many samples in order to make an estimate and where State spaces are large like the state spaces of all genomic diversity for an animal with a billion base pairs in its genome there's going to be certain things where it's just not really tractable in contrast variational methods are an information Theory driven approach that is going to use all the tools that will come to know and love like the KL Divergence and and the variational free energy so that's kind of like why we're taking this path at all because exact Bays would be epic but it's intractable exact base would allow us to compute the minimum surprise model which would be the maximum evidence model we're not going to be able to compute exact base in general how can we approximate it well with sampling and or variational sampling has some situational advantages but variational is a much much more flexible approach it's a lot more tractable it's incrementally optimizable we can reduce the KL Divergence and improve our model fit incrementally and kind of indubitably like we will always make the fit better if we reduce the kale Divergence we're making the fit better now that doesn't mean that the final fit or anywhere along the way is adequate for your business your company or you know whatever you could be moving a gaussian distribution to its best possible fit location in the middle of a bimodal distribution and still be totally missing the point so this isn't to say like it works it's just to say the ball rolls downhill that's like the big picture on why variational methods are being used so chapter four is going to describe

generative models which are using Bayesian networks using Bayesian methods to describe cognitive systems sense making an action and then the variational methods are going to support us doing optimization and inference with these models approaching exact base asymptotically but instead of focusing on surprise minimization itself um which we're not going to be able to directly calculate we're going to be able to calculate something that is bounding surprise and that's going to be the variational and in figure 2.4 we kind of saw what that looked like as well variational free energy equation 2.5 the distributions and the data and if we can shrink that Divergence we're getting ratcheting closer to surprise so that's why it's possible to use variational free energy to incrementally optimize and update and learn and use that um attractively Jensen's inequality would be awesome to have more of a um uh like an unpacking on this because this is kind of um it's very provocative in general but suffice to say that because of the curvature of the natural log function or any other function that's concave down like that were then able to use that property of this curvature type function and properties of logs in order to generate an inequality such that the log of the expectation is always greater than the expectation of the log so it's kind of like an application of Jensen's inequality for concave down functions in this setting heuristic bounding of what what we would really want to there's some further rearranging um again I'll review later perhaps Jonathan's um writing makes this uh clear or other people can fill in some dots um but where this is all kind of like more detail on equation 2.5 just helping us understand variational as kind of like the Bayesian optimal way to fit um where's Jonathan's writings uh on the chapter four page it is um here and I'll put it in the chat thank you Kate so that was 4.2 section 4.2 was just saying Bayesian inference frog jumping out of the hands if you can calculate it exactly go for it No One's Gonna Stop you but in situations where we want to do it approximatable heuristic and one that's incrementally optimizable we can use free energy this is also known as energy-based methods or energy-based learning or variational methods going back to decades ago so this is not invented by active inference at all it's also called the evidence lower Bound in Bayesian statistics and so this is just applying that methodology approximate Bayesian computation with a variational flavor not the sampling flavor in this case the question of perception you know having made the the software update from just pure exact base to approximate base now we can calculate the free energy for a generative model so whereas an exact base we would be talking about getting the maximum evidence model minimum surprise model plug and Chuck now we're going to get the minimum free energy which is going to be converging towards that minimum surprise maximum evidence um the

book is going to return and continue to juxtapose and contrast two major architectural decisions that are made for dynamical models now not all models are dynamical not all models involve something changing through time in general when you are modeling something that's changing Through Time you can either have a continuous time model or a discrete time model discretization sometimes still has to come into play for continuous models and so on but this is like the two broad approaches to dealing with dynamical systems continuous time discrete time and that's why figure 4.3 we're always coming back to again and again with the discrete time and with A continuous time because it's like a Rosetta Stone that's like saying whether you take this approach to modeling time or that there are some consequences and there's different settings where they may be more or less appropriate but we're still talking about the same infrastructure and logic for generative models but before building up to 4.3 which has this like downstairs perception passive perceptual inference component and then an upstairs decision making component and analogously for continuous time figure 4.2 is just getting us familiar with what these um causal graph structures are and do and so it's a fun exercise or question like to first off draw um different situations that you're interested in or thinking of as like Bayesian causal graphs and then also to look at the ones provided here and then think about what real world situations map onto these like maybe here's was there a lightning strike or not unobserved but then we observe the light and the sounds or there could be other examples but but this is just kind of like bring the level of rigor to causal flowchart mapping and these are really extensible and composable like you can connect them in many different ways and given their connectivity you've defined like the necessary and sufficient information for that model it's not like oh and there's a secret connection from you know this one back to that one it's like what's written is there and included and calculated with standard methods what's not included isn't included it'd be like a calculator it um is only going to add what you tell it's at so inference is going to proceed along this infrastructure without some other secret third thing influencing it so they they talk a bit more about just what what these um how to read the relationships between different variables then they get to figure 4.3 which is the Rosetta Stone figure between the discrete time and the continuous time um anyone want to give a thought or a question on figure 4.3 well we're going to return to the contrasts of discrete time and continuous time multiple times it's going to come back most fully in chapter 7 and chapter 8 which are wholly dedicated to discrete and continuous time but in this section it's going to foreshadow and discuss those two different types Darius presumably this model um is a kind of simplification of what might be going on

in the cognitive creature I because I presume there are some backward loops in which the observations that you make are informing your action policy hence the action perception Loop similarly I was also I also make the presumption that the state dependencies have to be informed by prior action policies and beliefs so priors about these so I presume it's just this architecture needs to be built out a little bit more or else it's not really governed by any particular Telos by any particular incentive yeah great points there is a prior that helps get the ball rolling um and and so that prior kind of starts the chain but then in subsequent um time steps the the action and the unfolding of the situation at the prior time Point um sets how things are and certainly these are like these are like uh like a frame it does not have more advanced cognitive features like attention anticipation so on those things get get layered in but this is kind of like the kernel or like The Motif that we then can can elaborate and bring richness to and so that's like why sometimes there's an adjective in front of active inference like sophisticated active inference or deep active inference affective active inference and those adjectives call attention to certain structural motifs that are being brought into play and this is kind of like this is like the lowest common denominator because we have a prior on hidden state was a prior on the temperature of the room and then we have the temperature of the room changing Through Time and the thermometer readings emitted at each time that's the downstairs passive inference bit D is a prior hidden State real temperature B describes how temperatures change Through Time a describes the relationship between temperature emitting observations or observations inferring back to temperature downstairs looking like an e facing down passive inference where do we embed action in The partially observable Markov decision process well the partially observable part is on the bottom because we have parts that are observable and parts that aren't so it's partially observable the markovian property through time is that the past only influences the future through the present partially observable Markov and then the last piece is the decision process P_i is your policy it's intervening in how the world changes Through Time and what guides policy selection well you could make a pomdp that has a stochastic policy selection or has reward driven policy selection or as we're going to see we utilize expected free energy to guide policy selection so this is kind of bringing the well-known pomdp format to bear on our setting Young yeah so I'm a bit confused about this generated model because as you said it's a generative model so do you it does this model actually mean that it's uh we're doing some kind of commenting process about this because um to what I understood is that uh number one number two and number three which is like the maybe the number three which should be notated as the transition metrics uh or a policy π_i is

actually something that is like a real dynamics of the external world not something that we are estimating so that's that's something that I'm being confused so because if you're saying generator model it means it sounds to me like it's something that is happening in our brain but in what I understood was that this is a diagram about the external World itself great great question so in fact we see two representations um one of them is the particular partition looking representation with the four internal and external blanket States here we don't see external States at all and it's not even shown like where they connect in and this is actually all happening within the um cognitive model of the agent I think the figure that that clarifies this the most is nine point one so now we're pulling back to the behavioral experimenters lab so here's going to be the rat in the maze we are using the pomdp um architecture to make a model of the rats beliefs about temperature changing through time and its beliefs about how policy influences that its beliefs about how temperature and thermometer are linked so the pomdp is describing the the agents parameterization it is not the actual true unfolding of the system here the observations are being actually the generative process is in this outer layer if we want to think about it that way um so yes the the B Matrix is something that the agent learns about how States change through time and how its actions intervene but it is that's not actually how things change Through Time that that that's tracked by another variable yeah oh yeah so does it mean that those uh EP function annotation key should be actually Q because it's an approximation isn't it so like uh I don't really remember which chapter it was but maybe chapter two the diagram actually kind of shows that he is actually a annotation for the notation for the uh probability happening in the real external World while two is something that we approximate so basically I think it's it's better it's much more accurate to say uh $Q_{\pi|q}$ given as yes I I would like to see someone else also um check on that however yeah I believe it makes sense to think about these as as the belief distributions of the agent and hence utilize Q whereas here p is just being used as like a generic statistical distribution but but I I am not 100 confident on that I I agree and that's that is how we think about it from the agent's perspective and I'll I I agree that the q's and the P's you know mind your q's and P's it does get confusing because it's like well wait but if we know anything about p why are we not just like using that information like why are we even messing around with Q if we can contrast it with with p so these like minor um notation ambiguity is actually making me kind of confused about the equations so I just wanted to clear well in the um I I would love to help connect the ontology terms which then we can support with translations and everything to the equations and to the difference because I I agree and the equations work out and then second layer renderings or

schematics often include or exclude things that that reduce the clarity of what they're representing um well in the rest of four discrete time is explored more that's basically more math about the top part of figure four three section four five continuous time more math on the bottom of figure four three some extremely um spicy and interesting topics with Markov blankets and message passing and some representations of message passing generalized coordinates of motion and some visualizations about that connecting it to Taylor series expansion LaPlace approximation another tool in the toolbox to help approximate a given inference problem by using a quadratic expansion around the mode more message passing and that's the end of four so definitely it's a dense chapter most of the density is in this second half of where it's just giving a lot more detail on discrete time and on continuous time and the big picture is we're introducing the kernel or the minimal Motif and some of the mechanics associated with using discrete time and continuous time models so next time we will turn to the cohort for questions so it'd be awesome if people add questions and and vote them and we'll come back and take it from any last thoughts okay thank you very well everybody bye